Diffraction

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Explain spreading around edges and through openings.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 3.*
24 m/s from $v=f\lambda$.

02. Worked example: explain the model
Strong curved spreading from narrow opening; mostly forward wavefronts with edge spreading from wide opening.

03. Read and interpret the model
Narrow opening produces strong spreading; wide opening mostly forward with edge effects.

04. Apply the physics
A, width comparable to λ.

05. Explain your reasoning
Edges/openings cause spreading within same medium; refraction involves speed change and boundary/gradient.

06. Review: connect an earlier idea *Review: Unit 5, Day 6.*
Same average per particle; total translational energy is 4 times as large.

07. Exit check
More, relative to the opening.

08. STRETCH: test the model
Usually less pronounced spreading when wavelength is smaller relative to opening.

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01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 3.*
v=30 m/s; λ=30/50≈0.6 m.

02. Worked example: explain the model
Strong curved spreading from narrow opening; mostly forward wavefronts with edge spreading from wide opening.

03. Read and interpret the model
Narrow opening produces strong spreading; wide opening mostly forward with edge effects.

04. Apply the physics
Sound λ1 m comparable to doorway; light λtiny relative opening, so far less noticeable diffraction at that scale.

05. Explain your reasoning
Edges/openings cause spreading within same medium; refraction involves speed change and boundary/gradient.

06. Review: connect an earlier idea *Review: Unit 5, Day 6.*
Rms speed becomes 4 times as large because mean kinetic energy is proportional to the square of rms speed.

07. Exit check
λhalves; less pronounced spreading when opening is large compared with new λ.

08. STRETCH: test the model
Wavelengths 2 and 0.2 m; 170 Hz has greater wavelength/opening ratio and stronger spreading.